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EDUCATION

Thiagarajar College of Engineering Madurai, Tamil Nadu
B.E. – Electronics and Communication Engineering **2023 – 2027**
CGPA: **8.86 / 10.0**

Sourashtra Girls Higher Secondary School Madurai, Tamil Nadu
Higher Secondary Certificate (HSC) **2022 – 2023**
Percentage: **96.83%**

Sourashtra Girls Higher Secondary School Madurai, Tamil Nadu
Secondary School Leaving Certificate (SSLC) **2020 – 2021**
Result: **PASS**

AREA OF INTEREST

Digital Electronics, Embedded Systems

TECHNICAL SKILLS

- **Programming:** C, Python
- **HDL:** Verilog
- **Embedded Systems:** Arduino Uno, ESP32, Xilinx ZCU104
- **EDA Tools:** Xilinx Vivado, Quartus II, MATLAB, LTspice

PROJECTS

FPGA–Based Low–Power YOLO Accelerator for Medical Sign Language Recognition — Xilinx ZCU104, YOLOv5
Team Project **Jul 2025 – Present**
- Developed an FPGA-based medical sign language recognition architecture using YOLOv5.
- Designed a smart power gating technique to reduce idle power consumption.

Edge AI–Based Driver Alertness System — Python, OpenCV, MediaPipe
Independent Project **Sep 2025 – Oct 2025**
- Developed an Edge AI system using OpenCV and MediaPipe to assess driver alertness through hand gestures.
- Measured reaction time and response accuracy to evaluate driver performance.

Smart Electrical Fault Detection and Public Safety Alert System — Arduino Uno, GSM Module, ACS712, Buzzer
Team Project **Aug 2024 – Dec 2024**
- Developed a system to detect abnormal current conditions in electrical lines using an ACS712 current sensor.
- Generated buzzer alerts and GSM notifications to improve public safety during electrical faults.

PROFESSIONAL EXPERIENCE

Quantanics Techserv Private Limited Madurai, Tamil Nadu
Embedded Systems Intern **June 2025**
- Developed ESP32-based sensor interfacing using the MQTT protocol and a Grafana monitoring dashboard.

CERTIFICATIONS

- **Digital Design with Verilog (Top 1%)** – NPTEL
- **Internet of Things (Silver Elite)** – NPTEL
- **SoC Post–Silicon Bring–up, Power & Performance Optimization** – NIT Trichy & Qualcomm (Workshop)
- **System Verilog & Semiconductor Design and Verification** – NIT Trichy & Synopsys (Workshop)

LEADERSHIP & ACTIVITIES

- **Office Bearer – Event Organizer, Electronics Hardware Club**: Planned and coordinated technical events.
- **Student Ambassador, NIT Trichy PROBE**: Supported technical outreach and workshop coordination.
- **Co–Chair, IEEE Solid–State Circuits Society (SSCS)**: Supported chapter activities and technical events.
- **Campus Ambassador, GUESSS India (IIT Mandi)**: Coordinated GUESSS India participation.
- **Achievements:** Academic Topper Award (2022), Prefinalist – Smart India Hackathon, Top 50 Nationally – HiDevs GenAI Skillathon, Ranked 74 out of 1670 – Naukri Campus TechQuezt #22